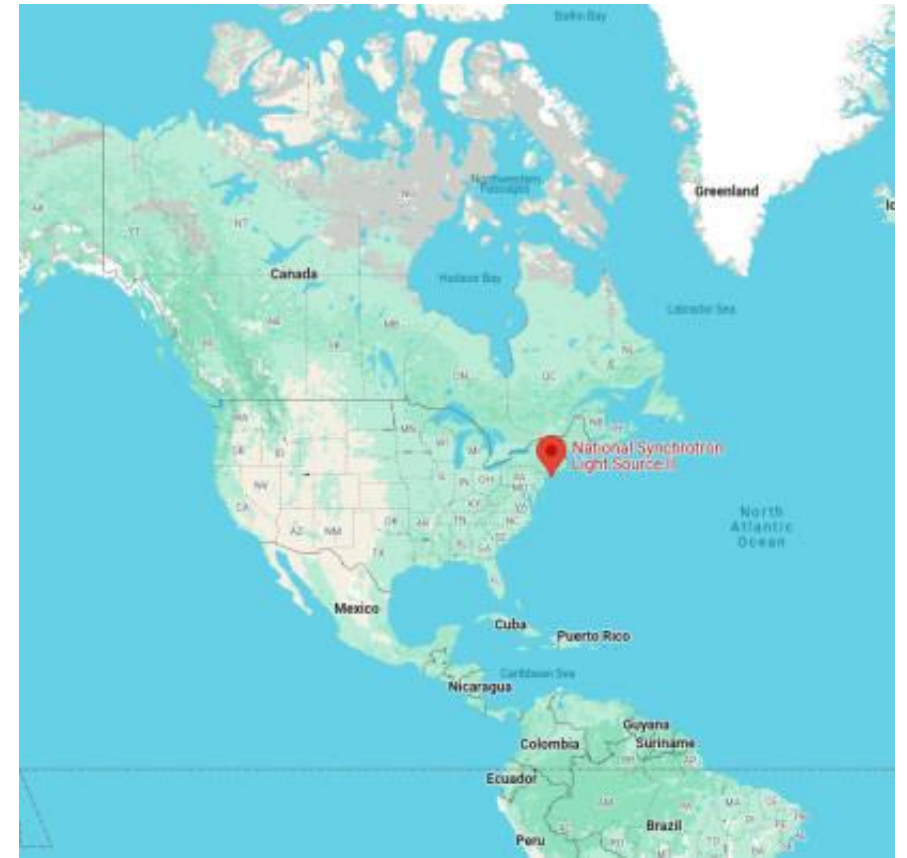


Facility Update – NSLS2 Motion Control

Robert Schaffer, Data Science and Systems Integration, NSLS2

Who are we?



National Synchrotron Light Source II

Status

- Mostly using TurboPMAC/PowerPMAC controllers
- Triggering coming from Zebra/Pizza/Panda box
- Flyscanning using Ophyd/Ophyd Async
 - Collaboration with DLS

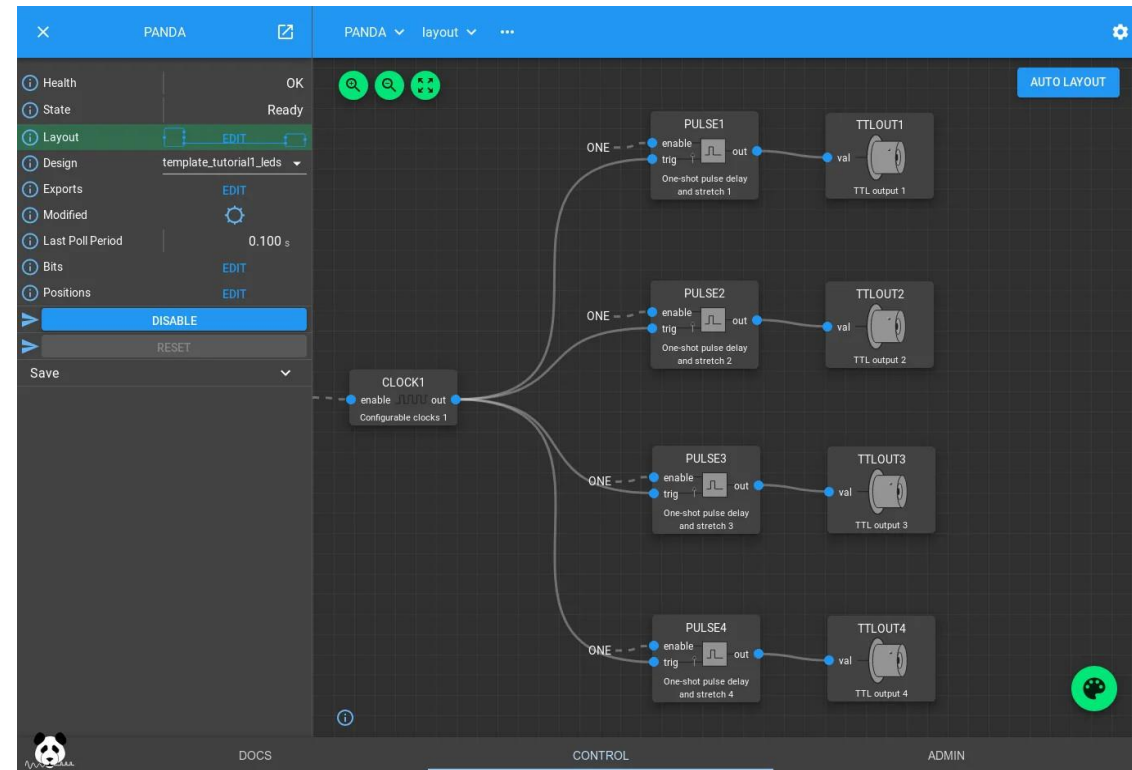
Controllers Status

- Facility standard is TurboPMAC and PowerPMAC controllers (Omron)
 - SDCC -> MCS2 transition for step scans
- Lots of others around
 - Aerotech
 - Galil
- Looking into MDrive intelligent motors
 - Onboard controls



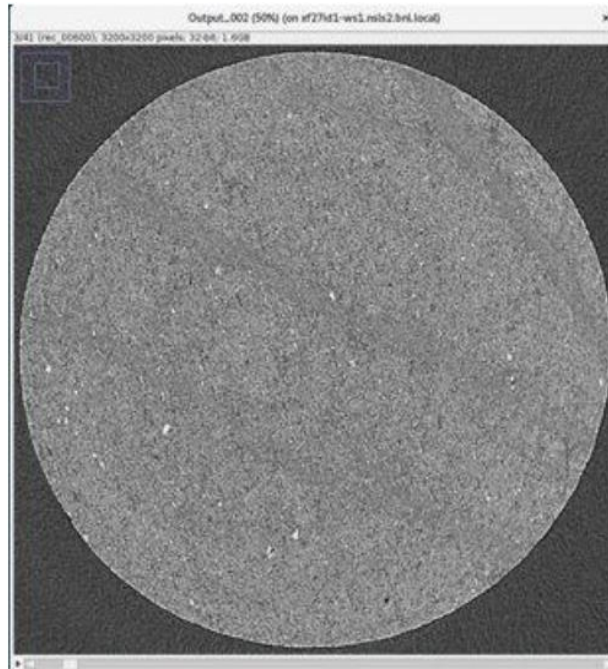
Signal Generators Status

- Moving toward PandaBox
 - FPGA
 - Successfully tested and integrated
 - Has GUI to configure board
- Lots of legacy hardware
 - Mostly ZebraBox
 - Some PizzaBox as well



Flyscanning

- Collaboration with DLS
- Tested for tomography at HEX



National Synchrotron Light Source II

Hardware and Software Stack

Planned transition from custom hardware and software-synchronized schemes to standardized hardware/software stack:

- Delta Tau Power PMAC motor controllers
- PandABox – FPGA-based scan coordinator
- Bluesky/Ophyd-async/Tiled – scan orchestration software
- EPICS – device control system

